

Reproducing FaceXFormer: A Unified Transformer for Multi-Task Facial Analysis

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Abstract

Unified multi-task models for facial analysis have emerged as a compelling alternative to task-specific pipelines, offering shared representations, reduced deployment cost, and real-time inference. FaceXFormer [5] is the state-of-the-art unified face analysis model, handling ten heterogeneous tasks within a single transformer-based encoder-decoder framework. Despite releasing inference code and pretrained weights, the authors did not release training code, dataloaders, or multi-task sampling infrastructure — making full reproduction a substantial and non-trivial engineering challenge. In this project, we undertake a principled, end-to-end reproduction of FaceXFormer across all ten tasks, rebuilding the entire training pipeline from scratch. We frame reproducibility itself as a scientific contribution: rigorously documenting every architectural assumption, every training decision left underspecified by the paper, and what the gap between claims and reproducible evidence reveals about publication norms in modern computer vision.

1. Proposal

What we plan to do. We propose a full reproduction of FaceXFormer [5], a unified transformer model for ten facial analysis tasks: face parsing, landmark detection, head pose estimation, attribute prediction, age estimation, gender and race estimation, facial expression recognition, face recognition, and face visibility prediction. The paper introduces a task-token-based encoder-decoder architecture that jointly processes all tasks via a novel bi-directional cross-attention mechanism. While the authors release an inference-only codebase with pretrained weights, the repository is critically incomplete for reproduction: it provides no training loop, no dataset loaders, no multi-task batch sampler, no loss implementations, and leaves key architectural branches undocumented (notably the hourglass landmark head, the PartialFC recognition integration, the exact segmentation

token-class mapping, and the unified head refinement path).

Our project reconstructs this missing infrastructure from scratch, guided by the paper, the released code, and principled engineering judgment. Concretely, we will: (1) audit the released repository and produce a structured paper-vs-code gap analysis for every component; (2) refactor the model into a modular, config-driven codebase with clean separation of backbone, MLP-Fusion, FaceX decoder, unified head, and all ten task heads; (3) implement dataset adapters and task-specific augmentation pipelines for all ten datasets; (4) implement and independently validate all loss functions; (5) build the task-balanced multi-task sampler; (6) train the full ten-task system on Purdue’s institutional GPU cluster; and (7) evaluate on all benchmarks reported in the paper, with explicit documentation of all assumptions made where the paper is underspecified.

This is not a reimplementing for its own sake. FaceXFormer embodies a precise set of architectural claims — bi-directional cross-attention over standard cross-attention, a 2-layer decoder over 9-layer alternatives, no face-specific pretraining — each supported by ablation results in the paper. Independent reproduction is the only way to validate these claims rigorously. We treat this project simultaneously as a systems engineering exercise and as a lens on the reproducibility problem in modern vision research.

2. Related Works

Task-specific face models. Historically, each facial analysis task has been addressed by a dedicated architecture optimized for that task alone. STARLoss [14] and Wing loss reduce semantic ambiguity in landmark detection; TokenHPE [12] uses transformer-based orientation tokens for head pose regression; ArcFace [3] defines the dominant paradigm for large-scale face recognition training via additive angular margin loss; DMUE and KTN advance facial expression recognition under label ambiguity. While these models achieve strong performance, they cannot be composed into a unified pipeline without redundant backbone forward passes, significantly increasing latency and mem-

ory overhead in any real deployment that requires multiple outputs simultaneously.

Multi-task face models. HyperFace [8] and AllinOne [9] are the pioneering convolution-based multi-task face systems, jointly handling detection, landmark localization, pose estimation, and gender recognition. More recent transformer-based multi-task models include SwinFace [6], QFace [10], and Faceptor [7]. These approaches adopt learnable task tokens inspired by DETR [2] but stop well short of full ten-task unification: QFace handles 4 tasks, SwinFace handles 4, and Faceptor handles 7. All rely on deeper (9-layer) decoders — Faceptor additionally uses a pixel decoder — leading to significantly higher computational cost and slower inference (14.30 FPS for Faceptor vs. 33.21 FPS for FaceXFormer). FaceXFormer [5] achieves this speed advantage through its 2-layer FaceX decoder with bi-directional cross-attention, without sacrificing accuracy.

Unified vision transformers. The broader push toward unified vision models — including SAM for segmentation, CLIP for vision-language alignment, and FaRL [13] for general facial representations — has demonstrated that shared representations across heterogeneous tasks produce more robust features than task-specific pretraining at equivalent model scale. FaceXFormer extends this philosophy to the ten-task face domain without large-scale face pretraining, relying instead on multi-task co-training across ten annotated datasets. Its MLP-Fusion module follows the SegFormer [11] design for efficient multi-scale feature aggregation.

Reproducibility in deep learning. A growing body of work has documented the systematic gap between published results and what independent reproducers can achieve, attributing failures to omitted hyperparameters, undocumented dataset splits, and missing training code. FaceXFormer is a canonical example of this problem: a high-impact, well-cited model with a public repository that is insufficient for training. Our project treats this gap as a first-class research object, not merely a practical inconvenience.

3. Approach

Architecture. We follow the FaceXFormer architecture as described in [5]. The encoder is a Swin-B transformer [4] pretrained on ImageNet, producing four-scale feature maps at strides 4, 8, 16, and 32. These are channel-projected to a common dimension and concatenated, then fused via a two-layer MLP-Fusion module (983K parameters) following the SegFormer design [11]. The resulting face representation $\mathbf{F}$ enters the FaceX decoder alongside learnable task tokens $\mathbf{T} = \langle T_1, \dots, T_n \rangle$. Each of the $N=2$ decoder blocks applies three operations sequentially: Task Self-Attention (TSA), where task tokens attend to each other to capture inter-task structure; Task-to-Face Cross-Attention (TFCA), where task tokens query face features to

gather task-relevant information; and Face-to-Task Cross-Attention (FTCA), where face tokens query task tokens to update the face representation with task-specific signal. The unified head applies one final TFCA before routing each token to its corresponding prediction head.

Token counts follow the paper: 19 for face parsing (one per semantic class), 68 for landmarks, 9 for head pose (one per element of the 3×3 rotation matrix), and 1 each for age, gender, race, expression, face recognition, visibility, and each of the 40 attribute labels.

Task heads and losses. Face parsing uses the refined face tokens $\tilde{\mathbf{F}}$ upsampled and cross-producted with parsing task tokens to produce a per-pixel logit map; loss is mean(Dice, CE). Landmarks use a hourglass network head on the 68 task tokens; loss is STARLoss [14]. Head pose uses a regression MLP mapping 9 tokens to a rotation matrix; loss is geodesic. Attributes and visibility use BCE with logits. Age uses mean(L1, CE) over decade bins. Gender and race use cross-entropy. Expression uses cross-entropy on 7 or 8 emotion classes. Face recognition uses PartialFC [1] with ArcFace loss [3]. The full joint objective across all ten tasks is:

$$\mathcal{L} = \lambda_{\text{seg}}\mathcal{L}_{\text{seg}} + \lambda_{\text{Ind}}\mathcal{L}_{\text{Ind}} + \lambda_{\text{hpe}}\mathcal{L}_{\text{hpe}} + \lambda_{\text{attr}}\mathcal{L}_{\text{attr}} + \lambda_{\text{age}}\mathcal{L}_{\text{age}} + \lambda_{\text{g/r}}\mathcal{L}_{\text{g/r}} + \lambda_{\text{exp}}\mathcal{L}_{\text{exp}} + \lambda_{\text{fr}}\mathcal{L}_{\text{fr}} + \lambda_{\text{vis}}\mathcal{L}_{\text{vis}} \quad (1)$$

Since the paper omits all λ_i values, we initialize them to 1.0 and apply gradient-norm balancing to determine task-specific scales, treating the final weights as reported hyperparameters in the paper.

Datasets. We use all ten datasets from the paper. For training: CelebAMask-HQ (face parsing, 30K images); 300W (landmarks, 3148 images, 68 keypoints); 300W-LP (head pose); CelebA (attributes, 202K images, 40 binary labels); UTKFace + FairFace (age/gender/race); RAF-DB + AffectNet (expression); MS1MV3 (face recognition, 5.1M images, 93K identities); COFW (visibility, 29 landmarks). For evaluation: CelebAMask-HQ, 300W full/common/challenge sets, BIWI, CelebA, LFWA, RAF-DB, UTKFace, FairFace, LFW, CFP-FP, AgeDB, CALFW, CPLFW, and COFW.

All images are resized to 224×224 . Landmark images receive keypoint alignment via five-point affine transform. Head pose images use loose crops to preserve orientation variability. Augmentation pipelines are implemented per Appendix F of [5]: random rotation ($\pm 18^\circ$), scaling ($\pm 10\%$), horizontal flip (50%), Gaussian blur, grayscale conversion, occlusion, and gamma adjustment — with per-task probabilities as specified.

Task-balanced sampling. We implement the upsampling-based multi-task sampler described in the paper. Smaller datasets are upsampled by repeating samples until all tasks have equal representation in each batch. Each batch of size 48-per-GPU across 8 GPUs draws samples from every ac-

tive task. We log per-task sample counts per epoch to monitor balance and detect drift.

Training. We train on Purdue’s institutional GPU cluster using 8 A100 GPUs with PyTorch DDP and FP16 mixed precision. The AdamW optimizer is used with initial learning rate 10^{-4} , weight decay 10^{-5} , decayed by $10\times$ at epochs 6 and 10, for 12 total epochs (plus 3 additional epochs for select tasks per the paper). Backbone weights are initialized from ImageNet-pretrained Swin-B. We checkpoint at each epoch and select based on held-out validation performance.

Staged training strategy. Given the scale of ten-task co-training, we adopt a staged approach to de-risk the experiment. We begin with 3-task co-training (parsing, landmarks, head pose) to validate the sampler and loss pipeline, then progressively expand to 6, 8, and finally 10 tasks. Each stage uses the checkpoint from the previous stage as initialization. This strategy also produces a natural set of ablation points — 3-task, 6-task, 8-task, and 10-task systems — that we can compare quantitatively.

Reproducibility methodology. At each component, we produce a structured gap analysis: what the paper claims, what the repository provides, what we infer, and the sensitivity of performance to that inference. This documentation is a primary output of the project alongside the model weights and evaluation results.

4. Anticipated Results

We structure our anticipated results across five experiment groups:

Baseline verification. We run the released checkpoint on all tasks and report metrics on standard benchmarks, confirming that the paper’s claimed numbers are achievable from the released weights and establishing our inference baseline.

Single-task baselines. We train each of the three core tasks (parsing, landmarks, head pose) independently to validate each loss implementation and task head in isolation before multi-task co-training begins.

Staged multi-task reproduction. We evaluate 3-task, 6-task, and full 10-task co-trained systems. Target metrics for the full system follow the paper: face parsing F1 ≈ 92.01 (CelebAMask-HQ); landmark NME ≈ 4.67 (300W full); head pose MAE ≈ 3.52 (BIWI); attribute accuracy $\approx 91.83\%$ (CelebA); age MAE ≈ 4.17 (UTKFace); visibility ≈ 72.56 recall@80%P (COFW); expression $\approx 88.24\%$ (RAF-DB); face recognition mean $\approx 95.94\%$ (LFW/CFP-FP/AgeDB/CALFW/CPLFW). We will report exact gaps between our numbers and the paper’s.

Ablations. We reproduce two ablations from the paper’s Table 7 under our independently trained model: (a) standard vs. bi-directional cross-attention in the FaceX decoder, and (b) balanced vs. unbalanced multi-task sampling. We addi-

tionally ablate loss weight initialization strategies ($\lambda_i = 1.0$ vs. gradient-norm balancing), which the paper leaves entirely unspecified.

Reproducibility analysis. We produce a structured table mapping each paper claim to: the degree to which the code release supports it, the assumptions required to implement it, and the measured performance gap. This analysis — covering architecture, losses, datasets, and training dynamics — is the distinctive contribution of this project relative to the original paper.

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